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Beyond membership inference: Distributional leakage in diffusion models☆

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ABSTRACT

Privacy concerns are growing as diffusion models are increasingly adopted in applications involving sensitive or regulated data. Membership inference attacks (MIAs) are commonly used to evaluate such risks, yet their effectiveness and underlying assumptions remain insufficiently understood, particularly with respect to reconstruction behavior and semantic conditioning mechanisms. Moreover, diffusion models learn rich semantic representations that may retain training-distribution information even when exact memorization is limited, suggesting that distributional leakage may emerge through mechanisms that are not fully captured by conventional attack-centric evaluation.

In this work, we present a privacy-auditing framework for diffusion models based on the notion of distributional leakage, combining retrieval-based analysis, reconstruction behavior, foreground-aware localization, and membership inference attacks under controlled settings. We introduce the concept of a reconstruction gap, defined as the difference in reconstruction fidelity between training-aligned and external samples, and investigate how it varies across diffusion architectures, conditioning strategies, and reconstruction settings. Our results reveal persistent distributional leakage across retrieval and reconstruction analyses, with generated samples systematically aligning with the training-aligned member distribution across domains, confidence tiers, and embedding spaces. We further show that attack effectiveness is strongly influenced by the spatial localization of semantic features, while explainability-based preprocessing preserves dataset-specific signals that enhance membership distinguishability.

1. Introduction

Diffusion-based generative models have rapidly become a central component of modern artificial intelligence (AI) systems, enabling high-quality synthetic content generation across domains such as healthcare, biometric processing, media synthesis, and data sharing (Yang et al., 2023; Hagos et al., 2024). As these models are increasingly trained on large-scale and potentially sensitive datasets, concerns regarding information leakage and unintended exposure of training information have grown substantially (Niu et al., 2025). In particular, the ability of generative models to learn rich semantic representations raises the possibility that sensitive information may be implicitly retained and later revealed through reconstruction processes or generated outputs.

Membership inference attacks (MIAs) (Barezzani, 2025c) have emerged as one of the primary approaches for evaluating privacy risks in machine learning (ML) systems (Duan et al., 2023; Zhang et al., 2024). By determining whether a target sample was part of a model's training set, MIAs provide a practical way to assess one manifestation of distributional leakage. We define *distributional leakage* as the presence of observable behavioral differences between member and non-member samples and study how such differences manifest through multiple complementary signals. Although extensively studied in discriminative settings, their behavior in diffusion-based generative architectures remains only partially understood. The iterative and stochastic nature of diffusion models, together with the increasing use of semantic conditioning mechanisms, complicates the analysis of how training information is encoded, reconstructed, and potentially exposed.

Existing mitigation strategies mainly aim to reduce the distinguishability between training and non-training samples. Differential privacy (DP) (Galende et al., 2025), for instance, introduces formal privacy guarantees by limiting the influence of individual training samples, often at the cost of generation quality and model utility. Other approaches, including regularization, data augmentation, and output perturbation (Ben Hamida et al., 2024; Jia et al., 2019), attempt to suppress exploitable signals and improve generalization (Barezzani et al., 2025). However, these methods generally treat privacy leakage as a uniform phenomenon, without explicitly accounting for the

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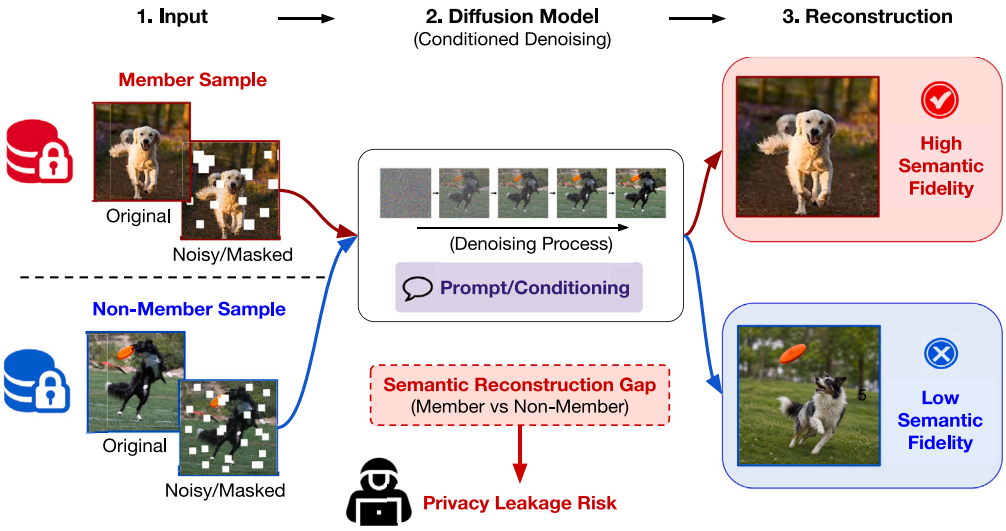


Fig. 1. Conceptual overview of distributional leakage in diffusion models. Training-distribution alignment may emerge through retrieval, reconstruction, and membership signals.

role of semantic content, reconstruction dynamics, or spatial feature representations.

In this work, we argue that information leakage in diffusion models cannot be fully characterized through attack-centric evaluation alone. Rather than focusing exclusively on attack performance, we analyze how diffusion models preferentially align with training-related data through retrieval behavior, reconstruction dynamics, and spatially localized semantic representations. Specifically, we investigate how distributional leakage emerges through retrieval proximity, reconstruction behavior, and membership distinguishability signals. To this end, we introduce the notion of a *reconstruction gap*, defined as the discrepancy in reconstruction fidelity between training-aligned and external samples. We study how this phenomenon evolves across different architectures, diffusion parameters, and conditioning strategies using both perceptual and pixel-level similarity metrics.

Fig. 1 illustrates the intuition behind our notion of distributional leakage. Given a partially observed input, diffusion models reconstruct missing content by relying on both the conditioning signal and the internal representations acquired during training. If the model has learned stronger priors for a specific data distribution, samples originating from that distribution are expected to be reconstructed with higher semantic fidelity than external samples. This discrepancy, which we refer to as the reconstruction gap, may create a measurable distinction between member and non-member data and therefore constitute a potential source of information leakage exploitable by attackers.

Building on the above analyses, we investigate whether privacy-relevant signals are spatially localized within generated content. In this regard, we evaluate membership inference attacks under multiple explainability-driven foreground extraction strategies, defining a fine-grained study of how semantic and structural cues contribute to membership distinguishability. This framework enables a more detailed understanding of how diffusion models encode semantic and structural information from their training distributions, and how such information can influence adversarial inference.

The main contributions of this work can be summarized as follows:

- We introduce a privacy-auditing framework for diffusion models based on the notion of distributional leakage, providing a unified perspective on retrieval, reconstruction, and membership-related privacy signals.
- We introduce and evaluate retrieval-based leakage indicators and the reconstruction gap, enabling the analysis of preferential alignment between generated outputs and training-aligned samples.
- We show that privacy-relevant information is retained primarily through semantic and structural representations rather than exact pixel-level memorization, revealing a persistent reconstruction bias toward training-aligned samples.
- We demonstrate that information leakage is inherently data-dependent and strongly influenced by semantic conditioning and the spatial localization of informative features.

Our experimental results reveal distributional leakage across retrieval, reconstruction, and foreground-aware analyses, indicating that diffusion models retain semantically meaningful information even when exact visual memorization is limited. These findings further show that information leakage is highly data-dependent, influenced by semantic conditioning and spatial feature localization, and therefore cannot be fully characterized through attack-centric evaluation alone.

The remainder of this paper is organized as follows. Section 2 reviews existing work on privacy risks and membership inference attacks in generative models. Section 3 presents the proposed privacy-auditing framework and formalizes the notions of distributional leakage and reconstruction gap. Section 4 provides a comprehensive experimental evaluation through membership inference attacks, retrieval-based analysis, reconstruction behavior, and foreground-aware localization. Finally, Section 5 summarizes the main findings and outlines future directions for privacy auditing in diffusion-based generative models.

2. Related work

The protection of the privacy of ML models is attracting growing interest. The initial focus of the research efforts has been the confidentiality of discriminative ML models, e.g., Shokri et al. (2017), Salem et al. (2019), and then increasing attention has been dedicated to generative ML models, e.g., Hilprecht et al. (2019), Chen et al. (2020). In either case, MIAs play an important role.

2.1. Membership inference attacks

The goal of a membership inference attack is to determine whether a given data sample was part of the training set of an ML model, raising critical privacy concerns (Barezzani, 2025c).

In the context of ML model confidentiality, MIAs are of particular interest from at least two different points of view.

On the one hand, MIAs do not necessarily require that attackers have detailed knowledge of the target ML model. As an example, in the black-box MIA described by Shokri et al. (2017), the attack is carried out by submitting a data sample and observing the model's output, relying on very limited additional information beyond the ability to query the target model. Therefore, considering the widespread availability of ML models on the web, MIAs represent a potentially pervasive threat to the privacy of ML models and, as such, could violate the requirements stated by the applicable regulations such as the European Union (EU) General Data Protection Regulation (GDPR) (European Parliament and Council of the European Union, 2016; Barezzani, 2025b) and Artificial Intelligence Act (AI Act) (European Parliament and Council of the European Union, 2024; Barezzani, 2025a). On the other hand, MIAs could be leveraged in privacy-assessment frameworks to evaluate the protection granted to personal data by the ML model considered (Ye et al., 2022).

The relevance of MIAs in the context of ML model confidentiality is further confirmed by the considerable number of attack techniques that have been proposed, developed and tested in centralized learning settings (Hu et al., 2022, 2023), as well as in federated learning (Bai et al., 2024).

Existing approaches can be broadly categorized into classifier-based and metric-based methods.

Classifier-based attacks train an auxiliary model to discriminate between members and non-members using features derived from model outputs. An example of such an approach is the already mentioned seminal paper from Shokri et al. (2017) that introduced a shadow model framework, showing that an adversary can exploit differences in model outputs between training and unseen samples to infer membership. The attacker model is trained as a binary classifier relying on the training dataset generated by means of the shadow models.

Metric-based attacks rely on statistical indicators. For example, Carlini et al. (2022) introduce the Likelihood Ratio Attack (LiRA), which relies on the TPR-at-low-FPR metric (where TPR is the True Positive Rate, and FPR stands for False Positive Rate). This approach allows us to take into account that, from a privacy perspective, not all the data samples are equal with respect to the confidentiality of the target ML model.

Such a consideration is of particular importance to avoid overlooking the role of data-specific features, such as embedding complexity, separability, or semantic uniqueness, which may significantly influence a sample's vulnerability.

2.2. Membership inference attacks in generative models

In general, the task assigned to generative models is more demanding than the task assigned to discriminative models. A generative model is required to learn the data distribution underlying the training dataset to an extent that allows the model to generate samples accordingly. MIAs have recently been extended to generative models, including generative adversarial networks (GANs) and variational autoencoders (VAEs) (Sun et al., 2023), as well as diffusion models (Tang et al., 2024). Since the target generative model is trained to approximate the underlying data distribution of the training set, the vulnerability to MIAs is an important aspect to consider. However, MIAs targeting generative models have peculiar characteristics. In fact, the absence of explicit class probabilities makes inference more challenging, especially in black-box settings where only generated outputs are observable.

Distance-based attacks (Sun et al., 2023) are based on the computation of the distance between the data sample to be tested and generated data samples. Although different distance metrics can be considered, the intuition underpinning these methods is the following: a generative model learns how to generate data samples from a proxy of the data distribution underlying the training data. Thus, if the data sample to be tested belongs to the training dataset, then its distance from data samples generated by the target model should be below a given threshold (Hilprecht et al., 2019).

Early attacks relied on reconstruction quality and similarity measures, showing that generative models tend to reproduce training samples more accurately than unseen ones. These approaches revealed that memorization effects can lead to privacy leakage even without direct access to model internals. More recent work has proposed architecture-specific attacks, such as exploiting discriminator signals in GANs or analyzing denoising trajectories in diffusion models. While some of these methods assume partial model access, there is a growing interest in black-box generative MIAs, where attackers rely exclusively on generated samples or output distributions.

It is worth mentioning that existing studies largely treat training data as homogeneous, focusing on average-case behavior rather than investigating how individual data characteristics affect leakage. In practice, certain samples, such as rare instances, outliers, or highly distinctive data points, are more likely to be memorized and thus more vulnerable to inference. This suggests that data distribution and feature-level properties are central to understanding privacy risks in generative models (Coscia et al., 2025), yet they remain insufficiently explored.

2.3. Membership inference defenses

Various defense strategies have also been proposed to mitigate MIAs by reducing the distinguishability between member and non-member samples (Hu et al., 2023).

Overfitting has been shown to be a sufficient, albeit not necessary, condition for MIA vulnerability (Shokri et al., 2017). Therefore, regularization-based methods which attempt to reduce overfitting are important membership inference defenses (MIDs). Techniques such as dropout, early stopping, and adversarial regularization improve generalization and reduce leakage.

A different approach is offered by data perturbation techniques. Differential privacy (Dwork, 2008) provides strong theoretical guarantees by limiting the influence of individual data points during training. However, these guarantees often come at the cost of reduced model utility, particularly in complex learning tasks. Other approaches focus on output-level defenses, such as confidence masking or adversarial perturbation (e.g., MemGuard (Jia et al., 2019)), which are particularly relevant in black-box settings, as well as privacy-preserving learning approaches that avoid the use of sensitive or correlated information during training (e.g., PriSM (Barbato et al., 2025)). While these methods aim to obscure informative signals exposed through model outputs, they may remain vulnerable to more advanced inference techniques.

Hybrid approaches combining data augmentation and distribution alignment have also been proposed to further minimize the gap between training and unseen samples.

The previously discussed risk of overlooking specific sample characteristics also applies to defense mechanisms. In fact, despite their diversity, most defenses are applied uniformly across all samples, without considering that privacy risk is inherently data-dependent. In particular, they do not explicitly account for how feature-level and content-specific properties influence the susceptibility of individual samples to membership inference, which can limit their effectiveness.

2.4. Privacy of diffusion models and explainable AI

Generative models have gained increasing importance due to the high quality of their generated outputs. One promising application of diffusion models is the generation of synthetic data which might be considered a potential solution to the trade-off between the demand for training data and the protection of personal information. A synthetic training dataset allows us to replace real data in the training dataset with generated data samples, preserving, to a given extent, the proximity of the obtained data distribution to the original one.

However, diffusion models are vulnerable to attacks that attempt to retrieve training data (Duan et al., 2023; Carlini et al., 2023). Therefore, synthetic data might not offer adequate privacy guarantees (Coscia et al., 2025).

It is therefore of paramount importance to further investigate the privacy risks entailed by the generation of data by means of diffusion models.

To this end, explainable AI (XAI) may offer an important contribution. XAI is employed to overcome the inherent opacity of ML models such as neural networks in an attempt to pursue AI trustworthiness (Ali et al., 2023). XAI also has a strong relationship with the vulnerability of ML models with respect to confidentiality attacks since it might leak useful information to an attacker (Liu et al., 2024). To develop a broader perspective on the privacy risks associated with diffusion models, we leverage the relationship between XAI and ML model confidentiality and use XAI as a tool to investigate privacy leakage with respect to semantic information.

3. Methodology

This section presents the proposed privacy-auditing framework for diffusion models. We first introduce the threat model and the formal definitions underlying our notion of distributional leakage. We then describe the construction of the semantically controlled benchmark and the three complementary analysis dimensions adopted throughout the paper.

3.1. Overview

While MIAs provide an effective framework for evaluating one manifestation of distributional leakage, they typically capture only a subset of the mechanisms through which diffusion models may retain and reproduce training data. In practice, leakage may emerge not only through membership distinguishability, but also through structural similarity, reconstruction fidelity, or reliance on semantically correlated visual patterns. To address these limitations, we adopt a broader leakage analysis framework organized into three complementary stages: (i) semantically controlled benchmark construction, (ii) retrieval-based leakage analysis, and (iii) reconstruction- and foreground-aware evaluation.

In the following, we first formalize the threat model and define the leakage indicators considered throughout the paper. Then, we describe the construction of semantically controlled member and non-member datasets and introduce three complementary analysis dimensions: retrieval-based leakage, reconstruction-based leakage, and spatially localized leakage.

To improve readability, Table 1 summarizes the main symbols used throughout the methodology.

3.2. Threat model and leakage definitions

We consider a diffusion model G trained on a private dataset D_{train} and deployed as a publicly accessible generative service.

The adversary does not have access to model parameters, gradients, optimizer states, or the original training data. Instead, the attacker can interact with the model through its public generation interfaces, including text-to-image generation and, when available, image-guided inpainting.

Let D_m and D_n denote the member and non-member datasets, respectively, and let $\mathcal{D}_m$ and $\mathcal{D}_n$ denote the corresponding underlying distributions. The dataset D_m serves as an empirical proxy for the unknown training distribution associated with $\mathcal{D}_{\text{train}}$.

For a candidate sample x, we define its membership label as

$$M(x) = \begin{cases} 1, & x \sim \mathcal{D}_m, \\ 0, & x \sim \mathcal{D}_n. \end{cases}$$

Table 1
Summary of the notation for each category.

Threat Model	
G	Diffusion model under evaluation
D_m	Member dataset
D_n	Non-member dataset
$\mathcal{D}_m$	Member distribution
$\mathcal{D}_n$	Non-member distribution
$M(x)$	Membership label of sample x
$S(x)$	Observable leakage score for sample x
Semantic Benchmark	
C_{target}	Target semantic class
$t(x)$	Semantic-confidence tier assigned to sample x
K	Number of classifier predictions considered
Retrieval Metrics	
$\mathcal{G}$	Set of generated samples
g	Generated sample
e	Retrieval encoder
$\Delta_t^{(1,e)}$	Nearest-neighbor similarity gap
$\Delta_t^{(k,e)}$	Top-k similarity gap
$R_{m,t}^{(e)}$	Member retrieval rate
k	Retrieval depth
Reconstruction and Localization	
Δ_R	Member/non-member reconstruction gap
m	Reconstruction mask
$\hat{x}$	Reconstructed image
$\Gamma(x)$	Class activation map (CAM)
$\mathcal{A}$	Allowed class set
τ_p	CAM percentile threshold
$h(x)$	Membership prediction for sample x

The adversary aims to infer information about $M(x)$ by exploiting observable behaviors of the diffusion model.

Unlike membership inference attacks that rely on training a binary membership classifier or on specific metrics, we adopt a broader privacy-auditing perspective. We assume that distributional leakage may manifest through multiple observable signals, including retrieval proximity, reconstruction fidelity, and spatial localization patterns.

The objective of our analysis is therefore not only to assess attack performance, but also to identify and quantify measurable indicators that reveal preferential alignment between generated outputs and the training distribution.

Definition 1 (*Distributional Leakage*)**.** Let $S : \mathcal{X} \to \mathbb{R}$ be an observable scalar score computed from the behavior of a diffusion model for an input sample x belonging to the image space $\mathcal{X}$. Let $P_m = \mathbb{P}(S(x) \mid x \sim \mathcal{D}_m)$ and $P_n = \mathbb{P}(S(x) \mid x \sim \mathcal{D}_n)$ be the corresponding score distributions. A diffusion model exhibits *distributional leakage* with respect to S if $P_m \neq P_n$.

In the definition above, the term *distributional leakage* refers to observable differences between the score distributions induced by member and non-member samples. The term *score* denotes any observable quantity derived from the model behavior and should not be confused with the score function used in score-based generative models. A discrepancy between P_m and P_n implies the presence of measurable membership-distinguishing signals. Therefore, an observable score satisfying Definition 1 may expose membership-related cues even when a specific MIA does not exploit it effectively.

In the following sections, retrieval similarities, reconstruction fidelity scores, and foreground-aware membership indicators are treated as concrete instances of $S(x)$. This allows our framework to characterize multiple leakage signals, rather than relying on a single attack model.

Definition 2 (*Reconstruction Gap*)**.** Let $R : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a reconstruction fidelity metric. Let $\hat{x}$ be the reconstruction of x produced

by the diffusion model. The *reconstruction gap* is defined as $\Delta_R = \mathbb{E}_{x\sim D_m}[R(x,\hat{x})] - \mathbb{E}_{x\sim D_n}[R(x,\hat{x})]$.

In the definition above, the *reconstruction gap* represents the difference in expected reconstruction fidelity between member and non-member samples. A positive value of Δ_R indicates that member samples are reconstructed more faithfully than non-member samples according to R.

The reconstruction gap is therefore a specific distributional leakage indicator obtained by choosing reconstruction fidelity as the observable score. In our experiments, we instantiate R using both structural and pixel-level metrics, enabling complementary views of reconstruction-based distributional leakage.

3.3. Member dataset construction

We now describe the benchmark construction procedure adopted throughout the experiments. The objective is to obtain semantically controlled member datasets for the leakage analyses presented in the following sections.

Let D_n denote a reference non-member dataset representing samples outside the target training distribution. The construction procedure described below focuses on obtaining semantically controlled member datasets D_m.

Let C_{target} denote a generic target semantic class. Our construction process combines semantic filtering, multimodal similarity estimation, adaptive confidence partitioning, and semantic verification to obtain datasets aligned with the target concept while minimizing ambiguity.

Given an image-caption pair $x_i = (I_i, T_i)$, we first retain samples whose visual or textual content is compatible with C_{target}. For each retained sample, we compute multimodal semantic consistency with respect to a high-confidence reference subset drawn from the non-member dataset D_n. Let e_i^{img} and e_i^{txt} denote normalized image and text embeddings, while $\mathcal{E}^{\text{img}}$ and $\mathcal{E}^{\text{txt}}$ represent embeddings extracted from the reference subset.

Semantic consistency is estimated through maximum cosine similarity:

$$s_i^{\text{img}} = \max_{e_j \in \mathcal{E}^{\text{img}}} \cos(e_i^{\text{img}}, e_j),$$

$$s_i^{\text{txt}} = \max_{e_j \in \mathcal{E}^{\text{txt}}} \cos(e_i^{\text{txt}}, e_j).$$

The final semantic consistency score combines visual and textual evidence:

$$s_i = \alpha s_i^{\text{txt}} + (1-\alpha) s_i^{\text{img}}.$$

Adaptive confidence partitioning is then performed using percentile-based thresholds computed over the score distribution. Given $S = \{s_i\}_{i=1}^N$, we define:

$$\tau_{\text{low}} = P_{60}(S), \qquad \tau_{\text{high}} = P_{80}(S).$$

Samples satisfying $s_i < \tau_{\text{low}}$ are discarded, while remaining samples are partitioned into hierarchical confidence subsets according to the adaptive thresholds. An additional classifier-based verification is applied to suppress ambiguous or semantically inconsistent samples.

Following Coscia et al. (2025), we additionally exploit a confidence-tier structure induced by an ImageNet classifier. Given the ordered top-K predictions $\pi_K(x) = (c_1, \dots, c_K)$ and a target-class set C_{target}, images are retained for tier assignment only if at least one of the top-K predictions belongs to C_{target} and the cumulative probability assigned to target classes exceeds a predefined confidence threshold.

For retained samples, we define the semantic-confidence tier as

$$t(x) = \max\left\{ r \in \{1,\dots,K\} \;\middle|\; c_i \in C_{\text{target}} \;\; \forall i \in \{1,\dots,r\} \right\}.$$

Thus, $t(x) = r$ indicates that the first r classifier predictions belong to C_{target}, whereas the $(r+1)$-th prediction does not, when $r < K$. If all top-K predictions belong to C_{target}, then $t(x) = K$. Higher tier values therefore correspond to stronger semantic agreement with the target concept. Samples for which the highest-ranked prediction does not belong to C_{target} are assigned to the auxiliary top-0 category, i.e., $t(x) = 0$. The resulting hierarchy therefore ranges from top-0 to top-K, with increasing tier values indicating stronger semantic agreement with the target concept.

This framework extends the methodology proposed in Coscia et al. (2025) beyond membership-oriented evaluation by providing semantically controlled member and non-member datasets suitable for fine-grained leakage analysis across retrieval similarity, reconstruction behavior, and foreground-aware evaluation.

3.4. Retrieval-based leakage

We introduce a multi-encoder, tier-aware retrieval pipeline designed to quantify potential training-data leakage in diffusion-based generative models. Our central hypothesis is that, if the model preferentially aligns with its training distribution, generated samples should lie systematically closer to the member distribution in feature space than to the non-member distribution.

To capture complementary semantic representations, we employ multiple frozen visual encoders and compute cosine similarity in the corresponding embedding spaces. For each generated sample g, encoder e, and semantic-confidence tier t, we retrieve the nearest neighbors from both the member and non-member subsets.

Rather than relying exclusively on the single nearest neighbor, we additionally consider the first k retrieved samples to improve robustness against retrieval noise and local feature-space variability.

Leakage is quantified through similarity gaps between member and non-member retrieval scores:

$$\Delta_t^{(1,e)}(g) = s_{m,t}^{(1,e)}(g) - s_{n,t}^{(1,e)}(g),$$

and:

$$\Delta_t^{(k,e)}(g) = s_{m,t}^{(k,e)}(g) - s_{n,t}^{(k,e)}(g),$$

where $s_{m,t}^{(1,e)}$ and $s_{n,t}^{(1,e)}$ denote nearest-neighbor similarities, while $s_{m,t}^{(k,e)}$ and $s_{n,t}^{(k,e)}$ represent average top-k similarities.

Positive values indicate that generated samples are systematically closer to the member distribution. We aggregate these scores over the full generated set to obtain dataset-level leakage estimates:

$$\bar{\Delta}_t^{(1,e)} = \frac{1}{|\mathcal{G}|} \sum_{g\in\mathcal{G}} \Delta_t^{(1,e)}(g),$$

$$\bar{\Delta}_t^{(k,e)} = \frac{1}{|\mathcal{G}|} \sum_{g\in\mathcal{G}} \Delta_t^{(k,e)}(g).$$

We additionally compute the member retrieval rate:

$$R_{m,t}^{(e)} = \frac{1}{|\mathcal{G}|} \sum_{g\in\mathcal{G}} \mathbf{1}\left[s_{m,t}^{(1,e)}(g) > s_{n,t}^{(1,e)}(g) \right],$$

which measures the fraction of generated samples whose nearest neighbor belongs to the member set.

Higher values indicate a stronger tendency for generated samples to retrieve member instances as their nearest neighbors, suggesting systematic alignment with the member distribution. However, positive similarity gaps may also arise from concentration around high-density regions of the training distribution. For these reasons, the proposed framework should be interpreted as measuring retrieval-based leakage indicators rather than providing direct evidence of exact memorization.

3.5. Reconstruction behavior

We design a controlled inpainting-based experiment to study how diffusion models reconstruct missing visual content and to what extent this behavior depends on the underlying data distribution. Our objective is to determine whether reconstruction fidelity differs between member and non-member datasets, thereby providing additional evidence of distributional alignment and potential privacy-relevant reconstruction effects. The intuition underlying this analysis is straightforward: if a generative model has acquired stronger prior knowledge about a specific image distribution, it should reconstruct missing regions from that distribution more faithfully.

Given an image $x \sim D_j, D_j \in \{D_m, D_n\}$, we define a binary mask $m \in \{0,1\}^{H\times W}$ that removes a predefined fraction of the image, producing the masked input $\tilde{x} = x \odot (1-m)$, where $\odot$ denotes the Hadamard (element-wise) product. A diffusion-based inpainting model G then reconstructs the missing region conditioned on a fixed prompt p, yielding the reconstructed image $\hat{x} = G(\tilde{x}, p)$.

To quantify reconstruction fidelity, we evaluate both structural and pixel-level similarity between the original image x and the reconstruction $\hat{x}$.

The generic reconstruction gap Δ_R introduced in Definition 2 is instantiated through two reconstruction fidelity functions. Specifically, we employ the Structural Similarity Index Measure (SSIM) (Wang et al., 2004) and the Peak Signal-to-Noise Ratio (PSNR). SSIM primarily captures perceptual and structural similarity, whereas PSNR is more sensitive to low-level reconstruction errors. The two metrics therefore provide complementary perspectives on reconstruction quality. These yield the corresponding leakage indicators Δ_{SSIM} and Δ_{PSNR}.

We evaluate both metrics globally over the entire image and locally over the masked region, i.e., by comparing x and $\hat{x}$ either directly or after restricting the comparison to the masked area through m.

To quantify dataset-dependent effects, we compute the reconstruction gap between member and non-member distributions:

$$\Delta_R = \mathbb{E}_{x\sim D_m}[R(x,\hat{x})] - \mathbb{E}_{x\sim D_n}[R(x,\hat{x})], \qquad R \in \{\text{SSIM}, \text{PSNR}\}.$$

Positive gaps indicate that member images are reconstructed more faithfully than non-member images, suggesting stronger alignment between the model and the member distribution.

3.6. Foreground isolation and background suppression

Background content can introduce dataset-specific biases, exposing membership information to attackers. To mitigate this effect, we design an XAI-based foreground-isolation pipeline aimed at suppressing background information while preserving semantically relevant objects.

Given an input image x, we compute a class activation map (CAM) (Zhou et al., 2016) using a pretrained ImageNet classifier. The resulting relevance map $\Gamma(x) \in [0,1]^{H\times W}$ highlights the image regions contributing most strongly to the classifier prediction. The relevance map is then converted into a binary foreground mask through a sequence of processing stages, including Gaussian smoothing, percentile-based thresholding, largest connected-component selection, morphological refinement, and optional GrabCut-based segmentation refinement. The final mask is then used to isolate the foreground object while suppressing background pixels.

A key component of the pipeline is the target-class selection strategy used to compute the CAM. Instead of relying exclusively on the classifier's top prediction, we introduce constrained target selection based on a predefined set of allowed classes $\mathcal{A} \subseteq \{1,\dots,C\}$, where C denotes the number of ImageNet classes.

Let $f(x) \in \mathbb{R}^C$ denote classifier logits and $\pi_K(x)$ the top-K predicted classes.

We consider the following target-selection strategies:

- *Top-1 Class.* The CAM is computed with respect to the highest-scoring class, $c^* = \arg\max_c f_c(x)$.
- *Constrained Top-1.* If an allowed class appears among the top-K predictions, the most confident allowed class is selected; otherwise, the method falls back to the standard top-1 prediction:

$$c^* = \begin{cases} \arg\max_{c\in\pi_K(x)\cap\mathcal{A}} f_c(x), & \text{if } \pi_K(x)\cap\mathcal{A} \neq \emptyset, \\ \arg\max_c f_c(x), & \text{otherwise.} \end{cases}$$

- *Class Aggregation.* The CAM objective aggregates the logits associated with all allowed classes, $\ell(x) = \sum_{c\in\mathcal{A}} f_c(x)$.
- *Fixed Class.* A predefined semantic class $c_0 \in \mathcal{A}$ is used as attribution target.

For all strategies, the CAM is normalized and thresholded according to:

$$\hat{\Gamma}(x) = \mathbf{1}\big(\Gamma(x) \geq \tau_p\big),$$

where τ_p denotes the p-th percentile of CAM activations. Higher percentile thresholds produce compact masks, whereas lower thresholds preserve larger foreground regions. The resulting mask is finally refined using connected-component analysis and morphological operations, optionally followed by GrabCut refinement.

The proposed design exposes several controllable factors. The target-selection strategy determines the semantic concept guiding attribution, directly influencing which image regions are preserved. Restricting attribution to $\mathcal{A}$ encourages attention toward semantically relevant categories, while aggregation produces more distributed activation patterns. The top-K parameter controls the permissiveness of the constraint by determining how many classifier predictions are inspected before falling back to the standard top-1 prediction when no allowed class is found. This parameter is referred to as the top-K fallback parameter. The percentile threshold regulates the trade-off between foreground completeness and background suppression.

The resulting foreground-isolated representations are subsequently used as input to a downstream membership inference classifier. Let $h : \mathcal{X} \to \{0,1\}$ denote a binary membership predictor operating on image space $\mathcal{X}$, where:

$$h(x) = \begin{cases} 1, & \text{if } x \text{ is predicted as member,} \\ 0, & \text{otherwise.} \end{cases}$$

Attack performance is evaluated using classification accuracy, the area under the receiver operating characteristic (ROC) curve (AUC), and the true positive rate at low false positive rates (TPR@lowFPR), following the evaluation protocol adopted in Carlini et al. (2022).

4. Experimental evaluation

This section presents a comprehensive evaluation of distributional leakage in diffusion models from four complementary perspectives: membership inference attacks, retrieval-based leakage, reconstruction behavior, and spatial localization of membership-related cues. While membership inference attacks provide a useful privacy baseline, they capture only a subset of the mechanisms through which diffusion models may retain information about their training distributions. Our goal is therefore to move beyond attack-centric evaluation and investigate whether diffusion models exhibit preferential alignment toward training-distribution samples through semantic retrieval proximity, reconstruction fidelity, and spatially localized semantic representations.

4.1. Experimental setup

This section summarizes the datasets, models, and evaluation settings adopted throughout the experimental evaluation. Unless otherwise specified, identical protocols are used across both semantic domains.

Datasets. We consider two semantically distinct domains, namely "vehicle" and "dog". Candidate member datasets are derived from LAION-Aesthetics v2 5+ (LAION, Large-scale Artificial Intelligence Open Network, 2022), while the non-member reference dataset is constructed from COCO (Lin et al., 2014).

For the dog domain, candidate member samples are constructed through the semantic filtering pipeline described in Section 3 and are organized into two confidence-based subsets, denoted as $\text{LAION}_{\text{Mid}}$ and $\text{LAION}_{\text{High}}$. For the vehicle domain, we instead adopt the LAION-0.50 and LAION-0.85 subsets introduced in Coscia et al. (2025). Although both pairs represent intermediate- and high-confidence member distributions, the naming convention differs because the two benchmarks are constructed through different dataset preparation procedures.

For the dog domain, a detector-assisted semantic filtering pipeline is employed. OWLv2 (Minderer et al., 2023) is used to identify candidate dog instances using prompts such as *dog*, *puppy*, and *pet dog*. Detected regions are refined through SigLIP (Zhai et al., 2023) similarity scoring. The final candidate score is computed as a weighted combination of the OWLv2 detection confidence, SigLIP similarity, and contextual consistency scores, using $(\lambda_1, \lambda_2, \lambda_3) = (0.60, 0.30, 0.10)$, respectively.

LAION candidate samples are filtered using ImageNet dog categories, an aesthetic-score threshold of 5, and multimodal CLIP similarity with $\alpha = 0.5$. Adaptive percentile thresholds corresponding to the 60th and 80th percentiles are used to construct the intermediate-confidence ($\text{LAION}_{\text{Mid}}$) and high-confidence ($\text{LAION}_{\text{High}}$) subsets.

Semantic confidence tiers. Both member and non-member datasets are partitioned according to the semantic-confidence tiers ($t(x)$) defined in Section 3, yielding the tier hierarchy from top-0 to top-5. This partitioning enables a fine-grained analysis of leakage as a function of semantic confidence.

Diffusion models. Stable Diffusion (SD) v1.4 (Rombach et al., 2022) is used as the primary generative model throughout the study. Additional experiments evaluate Stable Diffusion (SD) v1.5 (Rombach et al., 2022) and Stable Diffusion (SD) v2.0 (Stability AI, 2022) to assess cross-model robustness.

Generated samples. To reduce prompt-specific bias, we adopt a multi-prompt generation strategy composed of five semantically related prompts per domain. For each prompt, 1000 images are generated without fixing the random seed, resulting in a total of 5000 generated images per domain.

Retrieval pipeline. Retrieval-based distributional leakage indicators are evaluated using CLIP ViT-B/32 (Radford et al., 2021) and DINOv2-base (Oquab et al., 2024) embeddings. Similarity is measured through cosine similarity between ℓ_2-normalized embeddings. Both nearest-neighbor retrieval and top-k retrieval, with retrieval depth $k = 5$, are considered. For the within-member control described in Section 4.4, the same retrieval protocol is applied to two disjoint and size-matched partitions of the member benchmark. The partitions are stratified by member source and semantic-confidence tier and repeated over 10 random splits; within each split, the resulting indicators are macro-averaged across the five semantic-confidence tiers ($t = 1, \ldots, 5$).

Spatial localization analysis. To study spatially localized leakage signals, we evaluate several explainability-based foreground extraction techniques, including Rembg (Gatis, 2025), EigenCAM (Muhammad and Yeasin, 2020), GradCAM (Selvaraju et al., 2017), GradCAM++ (Chattopadhay et al., 2018), ScoreCAM (Wang et al., 2020), and LayerCAM (Jiang et al., 2021). CAMs are thresholded using the 80th percentile of activation values ($\tau_p = P_{80}$).

4.2. Baseline membership inference analysis

Before introducing the proposed leakage framework, we first establish a conventional membership inference baseline. This experiment serves two purposes. First, it provides a reference point against which the subsequent leakage indicators can be compared. Second, it allows us to evaluate whether attack-centric metrics alone are sufficient to characterize distributional leakage in diffusion models.

Fig. 2 illustrates representative examples from the four datasets used in the dog domain. Columns correspond to COCO, SD v1.4, $\text{LAION}_{\text{Mid}}$, and $\text{LAION}_{\text{High}}$, while rows represent classifier-induced confidence tiers. Several observations emerge. First, the semantic purity of the samples increases progressively across tiers, confirming the effectiveness of the filtering strategy. Second, the LAION subsets exhibit substantially stronger semantic alignment with the target class than the corresponding COCO reference images. Finally, despite this increasing semantic consistency, considerable visual diversity remains within each tier, highlighting the challenging nature of discriminating members from non-members based solely on visual appearance.

Table 2 reports the quantitative performance of the membership inference attacks. Two trends are particularly noteworthy. First, when the attack model is trained on Stable Diffusion images and evaluated against semantically filtered LAION subsets, performance degrades substantially. Although the attack remains moderately effective in terms of AUC, its effectiveness at very low false-positive rates collapses. This indicates that discriminating members from non-members becomes considerably more difficult when data distributions diverge despite semantic alignment.

Second, controlled settings reveal that the attack itself is not intrinsically weak. When training and testing are performed on Stable Diffusion versus COCO, near-perfect discrimination is achieved. Similarly, training and testing on the filtered LAION subsets yields high overall accuracy and AUC. These results demonstrate that the poor performance of the attack observed in realistic settings is primarily caused by the complexity and variability of the target distributions rather than by limitations of the attack model, as observed in Coscia et al. (2025).

The ROC curves shown in Fig. 3 further support this conclusion. While the controlled scenarios produce near-ideal ROC behavior, the realistic LAION settings exhibit substantially flatter curves and extremely low TPR values in the low-FPR regime.

These findings reveal an important limitation of conventional membership inference attacks: low attack effectiveness does not necessarily imply the absence of distributional leakage. More specifically, according to Definition 1, distributional leakage may still occur even when membership inference attacks achieve poor performance. In such cases, member and non-member samples can remain distinguishable through alternative observable scores, despite the inability of a particular attack model to effectively exploit these differences. This observation motivates the broader leakage analysis developed in the remainder of this section.

4.3. Retrieval-based leakage analysis

We now investigate whether generated samples exhibit systematic proximity to the candidate member distribution in feature space. Unlike membership inference attacks, retrieval analysis directly evaluates the relationship between generated images and training-aligned samples, providing a complementary view of memorization and leakage.

Table 3 reports retrieval-based leakage results across both domains and both embedding spaces.

Three observations emerge across all experiments.

First, all similarity gaps are positive, indicating that generated samples are systematically closer to candidate member samples than to non-members. This trend is stable across domains, tiers, and encoders,

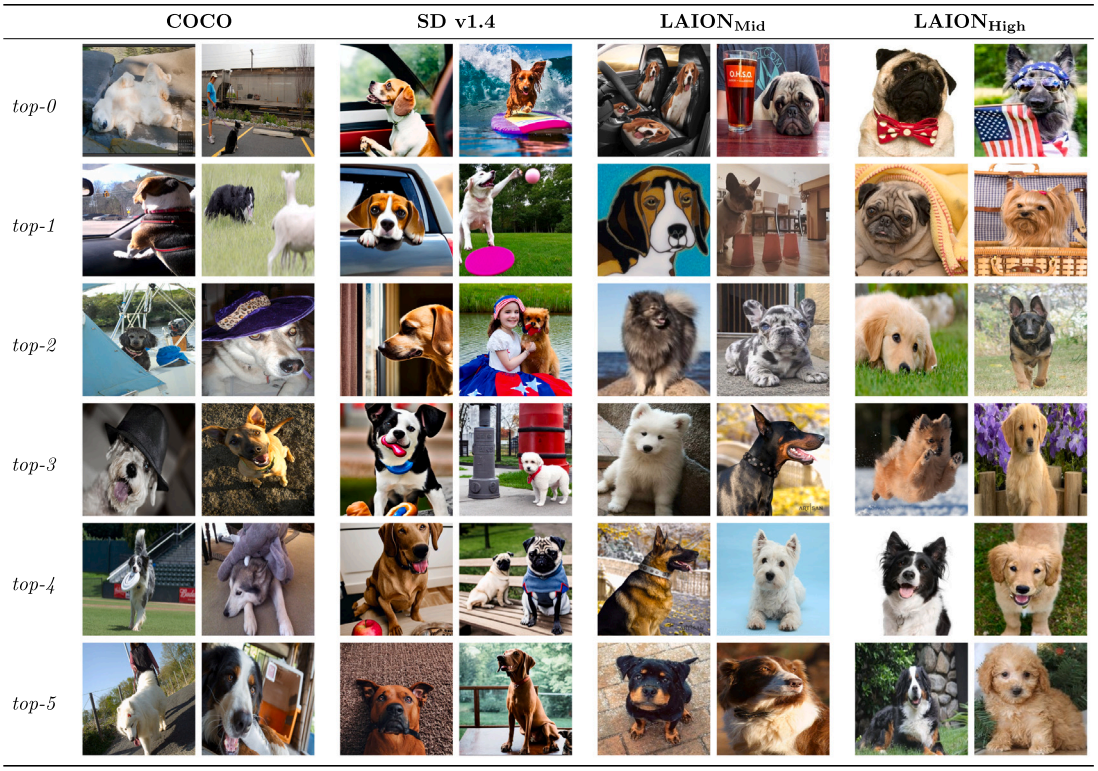
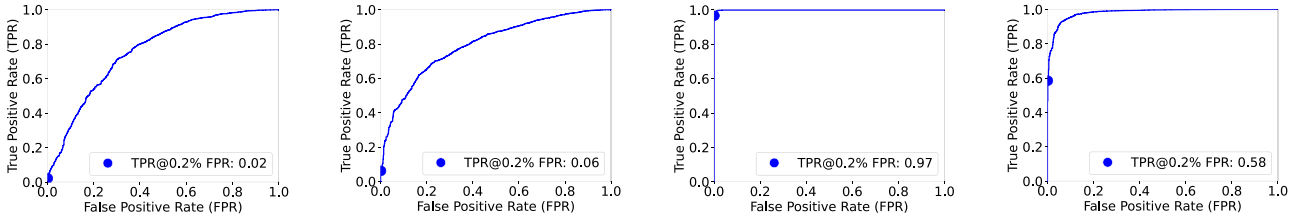


Fig. 2. Representative dog samples across datasets and semantic-confidence tiers.

Table 2
Membership inference attack performance under different training and testing settings.

Training images		Testing images		Evaluation metrics		
Non-Member	Member	Non-Member	Member	Accuracy	AUC	TPR@0.2% FPR
COCO	SD v1.4	COCO	LAION$_{Mid}$	74.77%	0.761	0.02
COCO	SD v1.4	COCO	LAION$_{High}$	53.26%	0.798	0.06
COCO	SD v1.4	COCO	SD v1.4	99.16%	0.999	0.97
COCO	LAION$_{Mid}$ LAION$_{High}$	COCO	LAION$_{Mid}$ LAION$_{High}$	96.90%	0.981	0.58



(a) SD v1.4/COCO → COCO/LAION$_{Mid}$ **(b)** SD v1.4/COCO → COCO/LAION$_{High}$ **(c)** SD v1.4/COCO → SD v1.4/COCO **(d)** LAION$_{Mid}$–LAION$_{High}$ → LAION$_{Mid}$–LAION$_{High}$

Fig. 3. ROC curves for cross-dataset and within-dataset membership inference experiments.

suggesting that retrieval-based leakage is a persistent property rather than an isolated artifact.

Second, CLIP and DINOv2 capture complementary aspects of leakage. CLIP generally produces higher member retrieval rates, indicating stronger semantic alignment between generated samples and the member distribution. DINOv2, in contrast, often yields larger similarity gaps, suggesting stronger instance-level correspondence.

Third, leakage persists across all semantic-confidence tiers. Although some variation is observed between tiers, the retrieval indicators do not exhibit a monotonic trend with increasing semantic confidence, while positive similarity gaps remain consistently present even for lower-confidence subsets. This indicates that retrieval leakage is not restricted to highly confident samples but instead affects the generated distribution more broadly.

Table 3
Retrieval-based leakage indicators across domains and semantic-confidence tiers. Positive gaps indicate preferential alignment with the member distribution.

Domain	Tier	No. of Images		CLIP			DINOv2		
		Members	Non-members	$\bar{\Delta}_t^{(1,e)}$	$\bar{\Delta}_t^{(k,e)}$	$R_{m,t}^{(e)}$	$\bar{\Delta}_t^{(1,e)}$	$\bar{\Delta}_t^{(k,e)}$	$R_{m,t}^{(e)}$
Vehicle	top-1	4156	1721	**0.080**	**0.086**	**0.914**	0.089	0.106	**0.757**
	top-2	3730	1017	0.059	0.073	0.887	0.090	0.106	0.742
	top-3	2754	752	0.052	0.071	0.788	0.092	**0.109**	0.731
	top-4	1069	645	0.070	0.083	0.819	**0.098**	0.108	0.694
	top-5	1096	1608	0.076	0.079	0.850	0.073	0.073	0.690
Dog	top-1	2626	532	0.051	0.066	0.963	0.076	0.105	0.859
	top-2	3030	409	**0.063**	0.074	**0.990**	0.082	0.106	0.860
	top-3	3555	333	0.040	0.059	0.961	0.084	0.103	0.888
	top-4	3147	273	**0.063**	**0.075**	0.978	**0.096**	**0.119**	**0.909**
	top-5	26 013	1806	0.038	0.047	0.967	0.071	0.085	0.899



(a) Vehicle Domain (b) Dog Domain

Fig. 4. Representative SD v1.4 generations used in the retrieval analysis. Summarized prompt categories are reported above each column.

The above findings provide evidence of persistent distributional leakage with respect to the retrieval-based scores introduced in Section 3. Positive similarity gaps $\bar{\Delta}_t^{(1,e)}$, $\bar{\Delta}_t^{(k,e)}$, and elevated member retrieval rates $R_{m,t}^{(e)}$ indicate systematic differences between member and non-member distributions, even when conventional membership inference attacks exhibit limited effectiveness.

Fig. 4 presents representative samples generated using the multi-prompt strategy adopted throughout the retrieval analysis. The examples demonstrate that the selected prompts generate visually diverse outputs while remaining aligned with the target semantic concept. This combination of diversity and semantic consistency helps mitigate prompt-specific biases and supports the interpretation of the observed retrieval effects as reflecting broader distributional alignment rather than artifacts of a particular prompt.

4.4. Within-member retrieval control

To assess whether the retrieval gaps observed in the main experiments could arise from the construction of the member benchmark itself, we introduce an additional within-member control.

For each domain, member source, and semantic-confidence tier, the corresponding member samples are randomly partitioned into two disjoint and size-matched subsets, denoted as $D_m^{(A)}$ and $D_m^{(B)}$. The split is performed independently within each stratum, thereby preserving both the member-source composition and the semantic-confidence composition of the original member distribution.

We then apply the same retrieval protocol adopted in Section 4.3, using the same generated Stable Diffusion samples as queries and the same CLIP ViT-B/32 and DINOv2-base embedding models. For a generated sample g, encoder e, semantic-confidence tier t, and retrieval depth k, we define the within-member gap as

$$\Delta_{A,B,t}^{(k,e)}(g) = s_{A,t}^{(k,e)}(g) - s_{B,t}^{(k,e)}(g).$$

As in the main retrieval analysis, we aggregate the per-sample gaps over the generated set as

$$\bar{\Delta}_{A,B,t}^{(k,e)} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \Delta_{A,B,t}^{(k,e)}(g).$$

Here, $k = 1$ denotes nearest-neighbor retrieval and $k = 5$ denotes the mean top-5 similarity. We additionally compute the corresponding preference rate:

$$R_{A,t}^{(e)} = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathbf{1}\left[s_{A,t}^{(1,e)}(g) > s_{B,t}^{(1,e)}(g) \right].$$

Since $D_m^{(A)}$ and $D_m^{(B)}$ originate from the same member distribution and are constructed through the same semantic-filtering procedure, no systematic preference toward either subset is expected. Accordingly, the within-member similarity gaps should approach zero and $R_{A,t}^{(e)}$ should approach the symmetric reference value of 0.5.

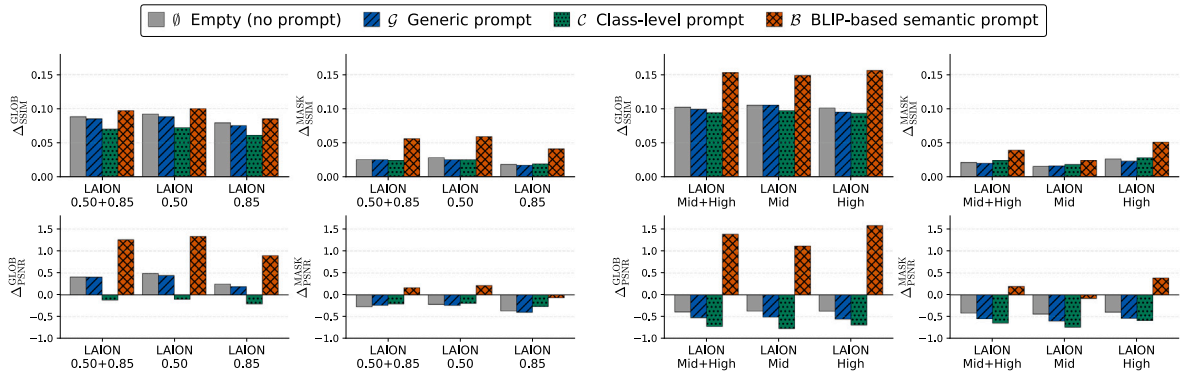
To reduce dependence on a particular random partition, the experiment is repeated over 10 independent stratified splits. For each split, the retrieval indicators are first computed independently for each of the five semantic-confidence tiers ($t = 1, \ldots, 5$) and then macro-averaged across tiers, assigning equal weight to each tier. Table 4 reports the mean and standard deviation of these macro-averaged indicators over the 10 random splits. For compactness, we denote the resulting tier-macro-averaged indicators as $\bar{\Delta}_{A,B}^{(k,e)}$ and $R_A^{(e)}$.

The within-member control exhibits the expected same-distribution behavior. Across both domains and both embedding models, the mean nearest-neighbor and top-5 similarity gaps remain close to zero, with values on the order of 10^{-3}. Moreover, the preference rate remains close to the symmetric reference value of 0.5, ranging from 0.479 to 0.511 across the considered settings.

These results indicate that randomly partitioning the semantically filtered member distribution into two size-matched subsets does not induce a systematic retrieval preference. In contrast, the member-vs-non-member experiments in Table 3 exhibit consistently positive retrieval gaps and substantially higher member retrieval rates. Therefore, the retrieval effects observed in the main experiments are not reproduced by sampling variability within the member distribution alone. This supports the interpretation that the observed differences are associated with preferential alignment with the training-related distribution rather than being solely induced by the construction of the member benchmark.

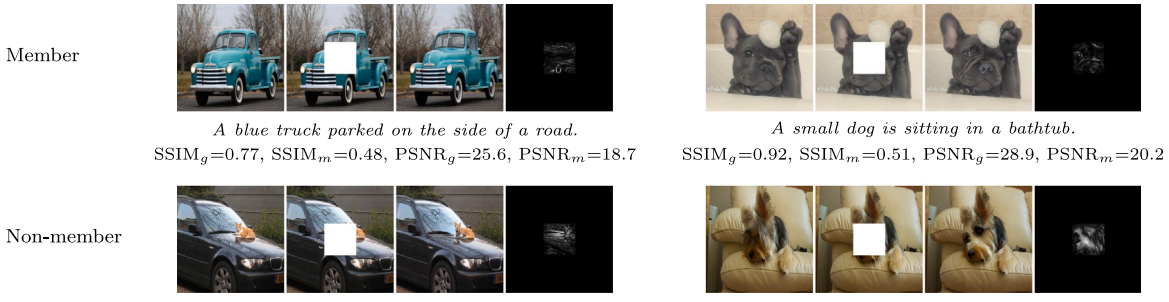
Table 4
Within-member retrieval gaps in units of 10^{-3}, macro-averaged across tiers; values denote mean ± standard deviation.

	CLIP			DINOv2		
Domain	$\bar{\Delta}^{(1)}_{A,B}$	$\bar{\Delta}^{(5)}_{A,B}$	R_A	$\bar{\Delta}^{(1)}_{A,B}$	$\bar{\Delta}^{(5)}_{A,B}$	R_A
Vehicle	0.8±1.8	0.7±1.6	0.479±0.034	0.6±3.9	0.1±3.9	0.489±0.028
Dog	1.2±3.1	0.4±2.4	0.511±0.066	0.4±2.0	0.1±1.9	0.494±0.015



(a) Vehicle Domain (b) Dog Domain

Fig. 5. Effect of prompt conditioning on reconstruction gaps for SD v1.4. Mask size and diffusion steps are fixed to 30% and 50, respectively.



A blue truck parked on the side of a road.
SSIM_g=0.77, SSIM_m=0.48, PSNR_g=25.6, PSNR_m=18.7

A small dog is sitting in a bathtub.
SSIM_g=0.92, SSIM_m=0.51, PSNR_g=28.9, PSNR_m=20.2

A cat sitting on the hood of a car.
SSIM_g=0.72, SSIM_m=0.24, PSNR_g=22.6, PSNR_m=13.9

A small dog lying on a pillow on a couch.
SSIM_g=0.78, SSIM_m=0.19, PSNR_g=23.9, PSNR_m=14.4

(a) Vehicle Domain (b) Dog Domain

Fig. 6. Representative inpainting reconstructions with BLIP prompts and reconstruction scores. Images are shown from left to right as original image, masked input, reconstruction, and reconstruction error map.

4.5. Reconstruction gap analysis

We next evaluate the reconstruction gap Δ_R introduced in Definition 2 and investigate whether diffusion models reconstruct training-aligned samples more faithfully than external images. To this end, we analyze reconstruction behavior through diffusion-based inpainting and quantify the resulting reconstruction gap.

Fig. 5 summarizes the effect of prompt conditioning on reconstruction behavior across the vehicle and dog domains.

In the vehicle domain (Fig. 5(a)), the reconstruction gap instantiated through SSIM remains positive across all datasets and prompt configurations, indicating that training-aligned samples are reconstructed with higher perceptual fidelity. However, under weak semantic conditioning (empty, generic, and class-level prompts), the PSNR-based gaps are generally close to zero or negative, particularly within the masked region. This suggests that the observed advantage is primarily structural rather than pixel-level. A markedly different behavior emerges when BLIP-based semantic prompts are employed. In this case, both global and masked reconstruction gaps increase substantially, yielding positive values for both SSIM and PSNR and revealing a stronger interaction between semantic conditioning and the model's learned priors.

A similar trend is observed in the dog domain (Fig. 5(b)), although the effect is generally more pronounced. Positive SSIM gaps persist across all prompt configurations and datasets, while weak conditioning typically produces negative PSNR gaps. Once image-derived semantic prompts are introduced, both global and masked reconstruction gaps increase considerably, with the largest improvements observed for the high-confidence member subsets. These results indicate that semantic conditioning amplifies the distinguishability between member and non-member samples, particularly for semantically coherent categories.

The positive values of Δ_{SSIM} support the distinguishability condition described in Definition 1, while the strong effect of BLIP-based prompts suggests that semantic conditioning can significantly amplify reconstruction-based leakage signals. These findings are consistent with the notion of distributional leakage introduced in Definition 1, as member and non-member samples remain detectable through reconstruction-based observable scores.

Fig. 6 provides representative qualitative reconstruction examples from the two semantic domains. For each sample, we report the corresponding BLIP-generated prompt together with the reconstruction fidelity scores.

While individual examples reveal visually meaningful differences between member-aligned and non-member samples, it remains important to determine whether such effects are specific to a particular diffusion architecture. We therefore extend the analysis to multiple

Table 5
Global and masked reconstruction gaps for different Stable Diffusion variants. Mask size and diffusion steps are set to 30% and 50, respectively. All reported values correspond to the gap between the indicated LAION subset and COCO.

Domain	Model	Dataset	Δ^{GLOB}_{SSIM}	Δ^{MASK}_{SSIM}	Δ^{GLOB}_{PSNR}	Δ^{MASK}_{PSNR}
Vehicle	SD v1.4	LAION-0.50+LAION-0.85	0.070	0.024	−0.119	−0.207
		LAION-0.50	0.072	0.025	−0.103	−0.196
		LAION-0.85	0.061	0.019	−0.207	−0.268
	SD v1.5	LAION-0.50+LAION-0.85	0.081	0.037	−0.086	−0.335
		LAION-0.50	0.083	0.040	−0.018	−0.273
		LAION-0.85	0.070	0.024	−0.326	−0.555
	SD v2.0	LAION-0.50+LAION-0.85	0.077	0.035	−0.061	−0.260
		LAION-0.50	0.079	0.037	−0.019	−0.172
		LAION-0.85	0.066	0.022	−0.316	−0.497
Dog	SD v1.4	$LAION_{Mid}$+$LAION_{High}$	0.094	0.024	−0.725	−0.648
		$LAION_{Mid}$	0.097	0.018	−0.774	−0.739
		$LAION_{High}$	0.093	0.028	−0.691	−0.587
	SD v1.5	$LAION_{Mid}$+$LAION_{High}$	0.153	0.035	1.213	−0.034
		$LAION_{Mid}$	0.149	0.019	0.969	−0.266
		$LAION_{High}$	0.157	0.046	1.413	0.143
	SD v2.0	$LAION_{Mid}$+$LAION_{High}$	0.151	0.035	1.427	0.264
		$LAION_{Mid}$	0.146	0.019	1.130	−0.021
		$LAION_{High}$	0.154	0.047	1.638	0.462

Stable Diffusion variants and evaluate the resulting reconstruction gaps across models.

In this regard, Table 5 evaluates whether the reconstruction gap remains stable across diffusion architectures. Positive SSIM gaps persist regardless of architecture, indicating that preferential reconstruction of training-aligned samples is not specific to a single model. At the same time, PSNR exhibits a more complex behavior, highlighting the distinction between perceptual similarity and pixel-level fidelity.

Interestingly, newer diffusion variants often amplify the reconstruction gap rather than suppress it. While these models generally improve perceptual reconstruction quality, they also appear to strengthen dataset-dependent reconstruction biases. This observation suggests that reconstruction leakage may represent a structural property of modern diffusion architectures.

The positive PSNR gaps observed for SD v1.5 and SD v2.0 in the dog domain suggest that semantic and pixel-level reconstruction fidelity may become aligned for highly structured categories, whereas the vehicle domain remains dominated by structural similarity effects.

4.6. Spatial localization of leakage

The previous analyses indicate that privacy-relevant information is retained at both semantic and reconstruction levels. We now investigate whether these signals can be spatially localized and exploited through explainability-driven foreground extraction techniques.

Table 6 reports attack performance obtained using different foreground extraction strategies. Some approaches preserve highly discriminative membership-related cues, whereas others suppress them almost entirely. More importantly, the ranking of methods differs depending on the evaluation metric, suggesting that different attribution strategies preserve different types of privacy-relevant information.

These findings reveal that distributional leakage is strongly associated with spatially localized semantic features rather than being uniformly distributed throughout the image. Membership-relevant signals appear to be concentrated in semantically meaningful regions that preserve discriminative information.

Fig. 7 provides qualitative examples of the foreground regions extracted by the various explainability techniques. The visual differences between methods are relevant. Generic background removal approaches preserve most of the object and surrounding context, whereas attribution-based techniques focus on specific discriminative regions. The correspondence between these localization patterns and the quantitative attack results suggests that membership-related signals are concentrated in semantically meaningful regions rather than uniformly distributed across the image.

Table 6
Membership inference performance obtained with different XAI-based foreground extraction methods.

Domain	Method	Attack performance		
		Accuracy	AUC	TPR@0.2% FPR
Vehicle	Rembg	73.27%	0.778	0.010
	EigenCAM	68.31%	0.790	0.060
	GradCAM	71.54%	0.723	0.008
	GradCAM++	70.17%	0.835	0.018
	ScoreCAM	75.56%	0.820	0.023
	LayerCAM	**79.25**%	**0.867**	**0.067**
Dog	Rembg	70.07%	0.764	0.010
	EigenCAM	88.31%	**0.905**	0.060
	GradCAM	**92.51**%	0.869	0.040
	GradCAM++	91.50%	0.848	0.010
	ScoreCAM	90.19%	0.877	0.030
	LayerCAM	66.42%	0.812	**0.090**

4.7. Cross-domain validation

Finally, we evaluate whether the leakage phenomena observed throughout the analysis generalize across semantically distinct domains.

Table 7 reports the complete cross-domain ablation study across vehicle and dog domains.

Evidence of distributional leakage is consistently observed across both vehicle and dog domains, indicating that the proposed framework captures a general phenomenon rather than a domain-specific artifact. Although the relative ranking of the evaluated methods varies across domains, leakage-related signals remain observable in both semantic settings.

The ablation study further reveals that attack effectiveness is influenced differently by the various components of the foreground-isolation pipeline. Restricting the set of allowed classes reduces performance in the dog domain, indicating that membership-related signals are distributed across multiple semantically related categories rather than being concentrated in a single subclass. Similarly, increasing the top-K fallback parameter does not yield monotonic improvements, suggesting that broader semantic coverage may introduce additional noise rather than stronger membership cues.

The sensitivity to CAM percentile thresholds highlights a trade-off between foreground completeness and localization precision. Different

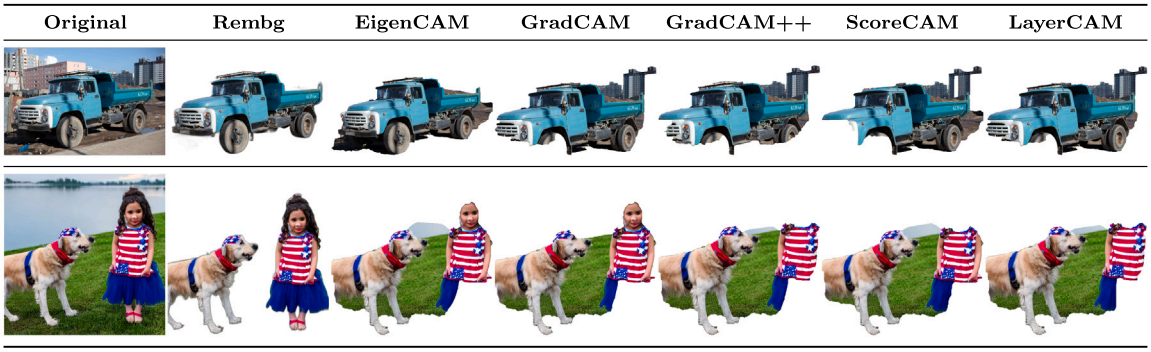


Fig. 7. Generated images (from SD v1.4) across background removal and attribution methods.

Table 7
Ablation study of the foreground-isolation pipeline for the vehicle and dog domains using LayerCAM. Best values within each factor are highlighted in bold.

		Vehicle domain			Dog domain		
Factor	Setting	Accuracy	AUC	TPR@0.2%FPR	Accuracy	AUC	TPR@0.2%FPR
Target Selection	Top-1	76.23%	**0.839**	0.047	97.18%	0.976	**0.230**
	Constrained Top-1	**77.00%**	0.819	0.015	**97.25%**	**0.979**	0.100
	Aggregation	71.52%	0.782	0.051	97.13%	**0.979**	0.130
	Fixed A[a]	73.16%	0.810	0.010	97.17%	0.975	0.080
	Fixed B[b]	76.31%	0.819	**0.065**	97.00%	0.975	0.090
Allowed Classes	All	**77.01%**	0.819	0.015	97.01%	**0.985**	**0.660**
	Subset 1[c]	75.29%	**0.841**	0.029	97.26%	0.983	0.240
	Subset 2[d]	72.49%	0.814	**0.087**	**97.54%**	**0.985**	0.240
Top-K Fallback	$K = 1$	71.44%	0.790	0.028	97.18%	0.982	0.440
	$K = 3$	**75.15%**	**0.865**	**0.074**	97.08%	**0.984**	**0.560**
	$K = 5$	71.51%	0.782	0.051	97.27%	0.983	0.470
	$K = 10$	71.92%	0.836	0.038	**97.40%**	0.982	0.140
CAM Percentile	60	69.19%	0.813	0.046	96.91%	0.979	0.200
	70	61.55%	0.835	**0.091**	97.05%	**0.983**	0.550
	80	73.29%	**0.851**	0.028	**97.58%**	0.982	0.110
	90	**76.83%**	0.813	0.030	97.10%	0.982	**0.570**

[a] Vehicle fixed class 705 (*Passenger car*) and Dog fixed class 207 (*Golden Retriever*).
[b] Vehicle fixed class 817 (*Sports car*) and Dog fixed class 235 (*German Shepherd*).
[c] Vehicle: car-like subset. Dog: small-dog subset.
[d] Vehicle: heavy-vehicle subset. Dog: large/working-dog subset.

threshold values preserve different types of privacy-relevant information, providing further evidence that leakage signals are spatially localized rather than uniformly distributed across the image. Consistent with the previous analyses, the dog domain generally shows stronger leakage signals than the vehicle domain, reinforcing the view that distributional leakage in diffusion models is fundamentally data-dependent.

The above results confirm the generality of the proposed framework and demonstrate that the observed leakage phenomena extend beyond a single dataset or semantic category.

Summary. The experimental results reveal that conventional membership inference attacks provide only a partial characterization of privacy risks. Retrieval proximity, reconstruction fidelity, and explainability-driven localization uncover complementary observable scores that satisfy the notion of distributional leakage introduced in Definition 1. These signals remain observable even when direct attack performance deteriorates. The consistent trends observed across retrieval-based, reconstruction-based, and foreground-aware analyses support the view that distributional leakage in diffusion models is a multifaceted phenomenon. However, since these analyses are evaluated on the same underlying member/non-member benchmark, their agreement should be interpreted as cross-method consistency across complementary analysis modalities rather than as statistically independent confirmation of leakage. In particular, systematic biases affecting the shared benchmark could influence multiple analyses simultaneously.

4.8. Threats to validity

Several factors should be considered when interpreting the results of this study.

Dataset validity. The LAION-based member datasets used in the main leakage analyses are derived from semantically filtered subsets of LAION-Aesthetics rather than from the original training datasets of the evaluated diffusion models. Consequently, the proposed member distributions should be interpreted as empirical proxies of the underlying training distribution.

Although the adopted filtering pipeline aims to maximize semantic consistency and reduce ambiguity, residual discrepancies between the selected subsets and the true training data may affect the magnitude of the measured leakage indicators.

Such distributional mismatches may also influence the effectiveness of membership inference attacks, as observed in the cross-dataset settings reported in Table 2. To assess whether the retrieval-based effects may instead arise from the construction of the member benchmark itself, we introduced the within-member control reported in Section 4.4. The near-zero similarity gaps and preference rates close to the symmetric reference value of 0.5 observed when comparing two matched partitions of the same member distribution indicate that the systematic retrieval preference reported in the main experiments is not reproduced by within-member sampling variability alone.

Cross-method dependence. The retrieval-based, reconstruction-based, and foreground-aware analyses probe different observable properties of the diffusion model, but they share the same underlying member/non-member benchmark. Consequently, agreement across these analyses should be interpreted as cross-method consistency rather than as statistically independent evidence of leakage. A systematic bias affecting the construction or composition of the shared datasets could potentially influence the different analyses in a similar way.

Model validity. Our evaluation focuses on three Stable Diffusion variants. While these models represent widely adopted diffusion architectures, the observed leakage patterns may not generalize to all generative models, including newer diffusion architectures, latent consistency models, or proprietary foundation models trained under different objectives and data curation strategies.

Retrieval and representation validity. The retrieval-based analysis relies on CLIP and DINOv2 embeddings to quantify similarity between generated samples and reference datasets. Although these encoders capture complementary semantic and visual information, different embedding spaces may emphasize different aspects of similarity. For this reason, the reported retrieval-based leakage indicators should be interpreted with respect to the chosen representation models.

Leakage interpretation. The proposed framework measures observable differences between member and non-member distributions through retrieval, reconstruction, and foreground-aware indicators. Positive leakage indicators therefore provide evidence of distributional alignment and membership distinguishability. However, such signals should not be interpreted as direct proof of exact memorization or verbatim reproduction of training samples. Instead, they characterize observable manifestations of distributional leakage that may increase privacy risks.

Despite these limitations, the consistent trends observed across multiple datasets, diffusion architectures, retrieval encoders, reconstruction settings, and foreground extraction strategies indicate that the reported leakage signals are reproducible across different experimental conditions.

5. Conclusions

We introduced a privacy-auditing framework for diffusion models based on distributional leakage. By combining membership inference attacks with retrieval, reconstruction, and foreground-aware analyses, the framework provides a broader perspective on privacy risks beyond attack-centric evaluation. Our results reveal distributional leakage, with generated samples aligning more closely with training-related distributions and positive reconstruction gaps indicating preferential reconstruction of training-aligned samples across architectures. The divergence between SSIM and PSNR suggests that leakage is more strongly reflected in semantic and structural similarity than in pixel-level reconstruction, while foreground-aware analyses show that it also depends on spatial localization of informative features. Importantly, the agreement across retrieval-based, reconstruction-based, and foreground-aware analyses should be interpreted as consistency across complementary analysis modalities rather than as statistically independent confirmation, since all rely on the same member/non-member benchmark.

Future work will extend the framework to emerging generative architectures and investigate mitigation strategies and privacy metrics beyond conventional membership inference attacks.

CRediT authorship contribution statement

Sergio Barezzani: Writing – review & editing. **Pasquale Coscia:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Stefano Ferrari:** Conceptualization. **Vincenzo Piuri:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

This study relies exclusively on publicly available data.

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